# Gepotidacin (Blujepa): From Novel Antibacterial Discovery to Clinical and Commercial Development

Bacterial infections remain an important global health concern, while the emergence and spread of antimicrobial resistance (AMR) threaten the effectiveness of existing antibiotics [1]. Resistance among pathogens responsible for urinary tract infections (UTIs) and sexually transmitted infections, including *Escherichia coli* and *Neisseria gonorrhoeae*, highlights the need for new antibacterial agents with novel mechanisms of action [2]. Furthermore, resistance to established treatments and limitations associated with their use can complicate treatment decisions and increase the need for effective alternative therapies [2].

Gepotidacin, marketed as BLUJEPA, is a novel oral antibacterial agent developed by GSK. It belongs to the triazaacenaphthylene class and inhibits bacterial DNA replication by targeting DNA gyrase and topoisomerase IV through a distinct mechanism from that of fluoroquinolones [2]. This mechanism has generated interest in gepotidacin as a treatment option for selected bacterial infections, particularly uncomplicated urinary tract infections and urogenital gonorrhea in eligible patients [2,3].

The clinical development of gepotidacin has included early-phase studies and Phase 3 trials evaluating its efficacy and safety [3,4]. These investigations have contributed to its regulatory evaluation and provide evidence for assessing its therapeutic value. However, its clinical role must also be considered in relation to adverse effects, emerging antimicrobial resistance, existing treatment options, and commercial considerations that may influence access and use [2–4].

This literature review aims to examine the discovery and development of gepotidacin, evaluate evidence from its clinical trials, discuss its therapeutic applications and safety profile, and critically assess antimicrobial resistance concerns and its potential market impact.

## 2. Drug Discovery and Development of Gepotidacin

## 2.1 Drug Discovery and Pharmacological Development

The development of gepotidacin was motivated by the growing need for new antibacterial agents as antimicrobial resistance (AMR) reduces the effectiveness of established treatments. Resistance among pathogens responsible for urinary tract infections (UTIs) and sexually transmitted infections has increased interest in antibiotics with novel mechanisms of action [1,2].

Gepotidacin, formerly known as GSK2140944, is a first-in-class antibacterial agent developed by GlaxoSmithKline (GSK). It belongs to the triazaacenaphthylene class and inhibits bacterial DNA replication by targeting DNA gyrase and topoisomerase

IV, enzymes essential for bacterial survival. Unlike fluoroquinolones, gepotidacin interacts with these enzymes through a distinct binding mechanism [2,5,6]. Its novel mechanism enables activity against selected bacterial isolates resistant to established antibiotic classes, although it does not eliminate the possibility of resistance developing.

Early pharmacokinetic studies evaluated oral absorption, systemic exposure, and urinary excretion. These properties were particularly relevant to uncomplicated UTIs because adequate urinary drug exposure is important for treating infections at this site. Findings from early clinical studies, together with microbiological and pharmacokinetic-pharmacodynamic evidence, supported further evaluation of gepotidacin and helped inform the dosing regimen used in later trials [7]. Thus, its development represented an effort to introduce a structurally novel oral antibiotic for infections where effective treatment options may be limited.

# 2.2 Preclinical Research and Antibacterial Activity

Preclinical research investigated gepotidacin's antibacterial activity using laboratory susceptibility testing, including minimum inhibitory concentration (MIC) measurements. These studies assessed its activity against clinically important Gram-positive and Gram-negative bacteria, including organisms associated with urinary and urogenital infections.

Biedenbach and colleagues reported in vitro activity against a broad range of bacterial pathogens, including selected isolates resistant to fluoroquinolones [5]. Mechanistic research further clarified how gepotidacin inhibits bacterial gyrase through interactions distinct from those of fluoroquinolones [6]. These findings supported its clinical development and suggested that its activity could extend to certain resistant bacterial populations.

Gepotidacin has also demonstrated activity against *Neisseria gonorrhoeae*, the bacterium responsible for gonorrhoea. This evidence provided a rationale for evaluating the drug as an oral treatment for uncomplicated urogenital gonorrhoea [2,4].

Nevertheless, laboratory activity does not independently establish clinical effectiveness. MIC results demonstrate inhibition under experimental conditions but cannot guarantee clinical cure, adequate drug exposure at every infection site, or prevention of resistance in patients. Preclinical findings therefore provided the scientific foundation for human studies, while clinical trials were required to establish efficacy and safety.

# 2.3 Clinical Development and Efficacy Across Trial Phases

### Phase I studies

Phase I studies evaluated gepotidacin's safety, tolerability, pharmacokinetics, and dose-related effects in humans. These early investigations characterized systemic exposure and urinary excretion, providing information needed to select doses for subsequent studies. Phase I research also identified safety considerations that required further evaluation during clinical development. Although early studies provide essential information about tolerability and drug disposition, they generally involve relatively small populations and cannot establish therapeutic efficacy on their own [2,7].

## Phase II studies

Phase II studies examined preliminary efficacy, safety, and dose selection in patients with bacterial infections. One Phase IIa study evaluated oral gepotidacin in 22 female participants with uncomplicated UTI. Participants received 1,500 mg twice daily for five days. At the test-of-cure assessment, 19 of 22 participants achieved clinical success; among the eight participants with qualifying baseline uropathogens, six achieved combined clinical and microbiological success [7]. These findings supported further investigation, although the small sample size limited the conclusions that could be drawn.

An earlier Phase II study also evaluated single-dose oral gepotidacin for uncomplicated urogenital gonorrhoea, providing preliminary evidence that supported further investigation in larger trials [10]. Phase II studies were therefore important for refining the clinical development programme, but confirmation in adequately powered Phase III trials remained necessary.

## Phase III trials

The EAGLE Phase III programme included trials evaluating gepotidacin for uncomplicated UTI and urogenital gonorrhoea.

EAGLE-2 and EAGLE-3 were randomised, controlled trials comparing oral gepotidacin, 1,500 mg twice daily for five days, with nitrofurantoin, 100 mg twice daily for five days, in eligible female participants with uncomplicated UTI. The primary outcome was therapeutic success, defined by combined clinical and microbiological response at the test-of-cure assessment [3].

In EAGLE-2, therapeutic success occurred in 50.6% of participants receiving gepotidacin and 47.0% receiving nitrofurantoin. In EAGLE-3, the corresponding rates were 58.5% and 43.6%, respectively. Gepotidacin met the prespecified non-inferiority criterion in both trials and demonstrated superiority to nitrofurantoin in EAGLE-3 [3]. These findings support its use for eligible patients with uncomplicated UTI caused by susceptible organisms. However, the results do not establish superiority over every available antibiotic or justify replacing established treatments in all patients.

EAGLE-1 evaluated gepotidacin for uncomplicated urogenital gonorrhoea. Participants received two oral 3,000 mg doses of gepotidacin, administered 10–12 hours apart, or the comparator regimen of intramuscular ceftriaxone plus oral azithromycin. Microbiological success was 92.6% with gepotidacin and 91.2% with the comparator, meeting the prespecified non-inferiority criterion [4]. The findings

support gepotidacin as an oral treatment option for eligible patients with gonorrhoea, particularly those with limited or no alternative treatment options.

Overall, clinical development progressed from early safety and pharmacokinetic investigations to confirmatory trials demonstrating efficacy for specific indications. Nevertheless, the trial populations, eligibility criteria, and follow-up periods should be considered when applying these findings to routine clinical practice [3,4].

# 2.4 Safety, Tolerability, and Therapeutic Applications

## Safety and tolerability

Gepotidacin's most frequently reported adverse effects include gastrointestinal symptoms, particularly diarrhoea and nausea. In the EAGLE UTI trials, diarrhoea was more common with gepotidacin than with nitrofurantoin, although reported events were generally mild or moderate [3]. In EAGLE-1, gastrointestinal adverse events also occurred more frequently in the gepotidacin group than in the comparator group [4].

The prescribing information includes important warnings concerning QTc prolongation, acetylcholinesterase inhibition and associated cholinergic effects, hypersensitivity reactions, and *Clostridioides difficile*-associated diarrhoea [8]. These risks should be considered when assessing patient suitability and reviewing concomitant medications. Continued safety monitoring is important because clinical trials cannot identify every uncommon adverse effect or fully predict outcomes in broader populations.

## Therapeutic applications

In the United States, gepotidacin is marketed as BLUJEPA. Its approved uses include uncomplicated UTI in eligible female adults and paediatric patients aged 12 years and older weighing at least 40 kg, when the infection is caused by specified susceptible organisms. It is also approved for uncomplicated urogenital gonorrhoea in eligible patients aged 12 years and older weighing at least 45 kg who have limited or no alternative treatment options [8].

Its oral formulation provides a treatment option for selected patients who may benefit from oral therapy. However, gepotidacin should not be regarded as a universal replacement for established antibiotics. Appropriate prescribing depends on the indication, susceptibility of the causative organism, local treatment guidelines, patient characteristics, contraindications, and available alternatives [3,4,8].

# 2.5 Antimicrobial Resistance and Limitations of the Evidence

Gepotidacin's novel mechanism of action does not make resistance impossible. Changes affecting DNA gyrase or topoisomerase IV can reduce susceptibility, and laboratory research has demonstrated that bacterial mutations can influence the

activity of gepotidacin [6]. The emergence and spread of resistant organisms therefore remain important concerns as clinical use expands.

Antimicrobial stewardship is essential to preserve the effectiveness of this antibiotic. Prescribers should confirm that treatment is appropriate, consider local resistance patterns, use susceptibility results when available, and avoid unnecessary antibiotic exposure. Surveillance of clinical isolates is also needed to identify changes in susceptibility and inform treatment recommendations.

Several limitations of the evidence should be acknowledged. In vitro studies cannot fully predict patient outcomes, and the Phase III trials enrolled defined patient populations under controlled conditions. Their findings may not apply equally to patients with complicated infections, severe comorbidities, or other characteristics that were not adequately represented in the studies. Furthermore, non-inferiority to a comparator does not establish superiority over all alternatives. Continued post-marketing safety monitoring and resistance surveillance are therefore necessary [3,4,8].

# 2.6 Market Impact and Commercial Significance

Gepotidacin addresses an unmet medical need by introducing a new class of oral antibacterial agent for selected urinary and urogenital infections. Its distinct mechanism and oral formulation differentiate it from established treatments, while its clinical evidence supports use in defined patient populations [2–4].

The commercial opportunity is influenced by the burden of UTIs, antimicrobial resistance, and the need for additional treatment options for gonorrhoea. Potential users include outpatient and sexual health services where oral treatment is appropriate. However, market uptake depends on regulatory approval, pricing, reimbursement, national formulary decisions, diagnostic practices, clinician confidence, and competition from existing therapies.

Antimicrobial stewardship may also influence commercial uptake because healthcare systems need to balance access to new antibiotics with the risk of resistance arising from inappropriate use. Consequently, clinical value and commercial success are related but distinct considerations.

GSK's 2025 annual report identifies BLUJEPA as a newly launched antibiotic following regulatory approvals in 2025 [9]. However, launch status alone does not establish long-term commercial success, and product-specific revenue should not be estimated without reliable disclosed financial data. Future market performance will depend on adoption, access, appropriate prescribing, continued clinical effectiveness, and the evolution of competing treatments [9].

In summary, gepotidacin has clinical and commercial potential because it introduces a novel antibacterial mechanism and expands treatment options for selected infections. Its long-term value will depend on maintaining effectiveness, demonstrating safety in broader use, and achieving appropriate access within antimicrobial stewardship programmes.

# 3. Conclusion

Gepotidacin (BLUJEPA) represents an important advancement in oral antibacterial therapy. As a first-in-class triazaacenaphthylene antibiotic, it inhibits bacterial DNA replication by targeting DNA gyrase and topoisomerase IV, offering a novel mechanism against selected bacterial pathogens.

Clinical evidence supports its use for defined indications. The Phase III EAGLE-2 and EAGLE-3 trials demonstrated non-inferiority to nitrofurantoin for uncomplicated urinary tract infections, while EAGLE-1 demonstrated non-inferiority to ceftriaxone plus azithromycin for uncomplicated urogenital gonorrhoea [3,4]. However, gastrointestinal adverse effects are common, and precautions regarding QTc prolongation, drug interactions, and other potential adverse reactions remain important [8].

Despite its potential, limitations include restricted evidence in certain patient populations and the possibility of emerging bacterial resistance. Continued surveillance, appropriate patient selection, and antimicrobial stewardship are therefore essential to preserve its effectiveness.

Commercially, gepotidacin offers a new oral treatment option for infections with limited suitable alternatives. Nevertheless, its long-term clinical and market impact will depend on accessibility, affordability, post-marketing safety evidence, and responsible prescribing.

In conclusion, gepotidacin is a promising addition to existing antibacterial treatments, but it should complement rather than universally replace established therapies. Further clinical experience and resistance monitoring will help define its future role in practice.

# Referenece


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